

17. Perform Edge detection using Sobel Matrix along X axis.

AIM:

Edge detection using Sobel Matrix along X axis.

PROGRAM:

```
import cv2

img = cv2.imread(r'C:\Users\Susmitha\Downloads\ITA0510 Lab\expl.png', 0)

if img is None:
    print("Error: Image not found!") else:
    # Apply Sobel filter in the X direction    sobel_x =
    cv2.Sobel(img, cv2.CV_8U, 1, 0, ksize=5)

    # Save the output image    cv2.imwrite(
    r'C:\Users\Susmitha\Downloads\ITA0510 Lab\sobel_x.jpg',
    sobel_x
    )

    # Display the output
    cv2.imshow("Original Image", img)
    cv2.imshow("Sobel X", sobel_x)

    cv2.waitKey(0)
    cv2.destroyAllWindows() INPUT:
```



OUTPUT:



RESULT:

Thus the program is executed successfully.